HCIS 342	Health Information Systems II	3(3,0)
In this course, students explore current technologies, regulations, and standards, including picture archiving and communication systems (PACS); the Health Insurance Portability and Accountability Act (HIPAA); 21CFR Part 11; FDA General Principles of Software Validation; and Health Level Seven (HL7), and examine their effects on software development. Other topics include information technologies for storing and maintaining data quality, ensuring safety, enforcing security, and electronic health record systems. **Prerequisite:** HCIS 341.		

HCIS 343	Information Security and Privacy in Healthcare	3(3,0)
This course introduces students to emerging issues in healthcare information security, privacy, and regulatory compliance. It also provides the students with a substantive overview and analysis of relevant information security subject matter that directly and materially impacts the healthcare system. **Prerequisite**: HCIS 341.		

HCIS 399	Co-op Training Experience	1(1,0)
Each student must complete 8 weeks of practical training in a field related to their interests. This Co-op work experience is usually fulfilled during the summer semester of the third year in the program. Students must submit a formal report and poster and make a formal presentation about their Co-op experience. **Prerequisites**: ENGL 217; Senior Standing.		

HCIS 444	Healthcare Business Intelligence and Data Analysis	3(3,0)
This course enables students to learn how Healthcare Analytics and Health Information Exchange (HIE) solutions can help uncover new growth opportunities and gain an advantage in the profitable healthcare technology market. Other areas covered include providing quality patient care while complying with government regulations, controlling costs, and increasing your organization's visibility in a crowded healthcare field. **Prerequisite**: HCIS 341.		

HCIS 445	Research in Healthcare Information Systems	1(1,0)
This course focuses on current research trends in healthcare information systems. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite:** Senior Standing.		

HCIS 490	Summative Learning Experience	3(3,0)
This is a graduation project directed at individual problems and research. Each student must complete it in their emphasis area. **Prerequisite:** ENGL 217.		

Business Required Courses

As described below, the HCIS program requires two business courses, BMKA 200 and BMGT 300.

BMGT 300	Project Management	3(3,0)
This course provides students with the skills to manage their business projects using effective techniques for leading, organizing, scheduling, and controlling tasks that advance project goals. Topics include project selection and statement of work; project managers' skills and task breakdown structure; PERT/CPM scheduling and budgeting. **Prerequisite:** Junior Standing.		

BMKA 200	Introduction to Marketing	3(3,0)
This course introduces the basic principles, theories, and marketing practices in our modern, ever-changing business environment. The course covers the marketing process activities of creating value for customers and capturing value from customers in return. It also discusses the marketing mix and how to build long-term customer relationships. Students will analyze case studies about a “real-life” product or service. Videos and in-class discussions on current marketing topics will assist in the learning experience. **Co-requisite**: ENGL 210		

II. Elective Courses

Descriptions of the major elective courses are given below.

COSC 434	Advanced Web Programming	3(3,0)
This course builds on the web programming course and prepares students to become full-stack developers. It allows students to learn how to develop back-end programs, connect their website or web application to a database, use regular expressions, implement asynchronous client access to server data, work with structured data formats such as JSON, and work with third-party media like web services. **Prerequisites:** COSC 231 and COSC 333. Equivalent to CCEE 514		

COSC 435	Mobile Application Development	3(2,2)
This course examines the principles of mobile application design, development, and testing. It covers memory management, user interface design, input methods, data handling, and network techniques. Students are expected to work on a project that produces a professional-quality mobile application. Projects will be deployed in real-world applications. **Prerequisite**: COSC 214, Equivalent to CCEE 511		

COSC 436	Game Programming	3(2,2)
This course provides an overview of software technologies for game development. It gives students a conceptual understanding of game design and practical experience in game creation. **Prerequisites**: COSC 214, Senior standing.		

COSC 438	Introduction to Software Testing	3(3,0)
This course will introduce students to software testing, its importance, goals, and approaches, as well as the roles of the major players (developers, testers, and users). Students will learn about the software testing life cycle, standards, types of testing (manual vs. automated), testing methods (black-box vs. white-box), testing levels (functional, unit testing…), and documentation. This course will also provide hands-on experience with automated testing tools. **Prerequisite***:* COSC 341.		

COSC 451	Operating Systems	3(3,0)
Students learn the concepts of operating systems theory and their implementations. Covered topics include computer and operating system structures, process and thread management, process synchronization, deadlocks, security and protection, inter-process communication, memory management, and I/O control systems. A case study of a UNIX operating system will accompany the course. **Prerequisite:** COSC 214. Equivalent to CCEE 513.		

COSC 460L	Networking Lab 2	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications (CCNA 1). The student will learn the functionalities of network equipment and protocols, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts through hands-on skills using hardware devices. **Co-requisite**: COSC 360L or CCEE 354L. Equivalent to CCEE 554L.		

COSC 461	Advanced Networking	3(3,0)
This course prepares students to act as a System and Network Administrator by implementing Active Directory Domain Services (ADDS in distributed environments that can include complex network services and domain controllers. The covered materials help students efficiently automate the administration of users, groups, and computers. **Prerequisite**: COSC 360 or CCEE 354.		

COSC 461L	Networking Lab 3	1(0,2)
This Lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Prerequisite**: COSC 460L or CCEE 554L. Equivalent to CCEE 555L..		

COSC 462L	Networking Lab 4	1(0,2)
This lab helps prepare students seeking to pass the Cisco Routing and Switching Essentials (CCNA 3) Certification. The primary focus of this Lab is routers and switches in large and complex networks. The student will learn how to configure routers and switches with advanced functionalities. They will also learn how to configure and troubleshoot enhanced switching technologies, first-hop redundancy protocols in a switched network, wireless routers and clients, and routers in complex networks. Finally, they will learn to manage CISCO IOS software licensing and configuration files. **Co-requisite**: COSC 461L or CCEE 555L. Equivalent to CCEE 556L.		

COSC 463	Cybersecurity operations	3(3,0)
This hands-on, career-oriented course focuses on practical topics to prepare students to acquire the required skills to work in the cybersecurity domain. The course will help students seeking to pass the Cisco Cybersecurity Operations Associate certification. **Pre-requisite**: COSC 360 (or CCEE 354).		

COSC 481	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include Intelligent agents, informed and uninformed search strategies, and adversarial search. The Python language libraries will also be introduced. **Prerequisite**: COSC 214. Equivalent to CCEE 562.		

COSC 482	Data Science and Web Scraping	3(3,0)
Data is becoming the fuel of the 21st century, and acquiring data processing and analysis skills is becoming necessary. This course introduces data science processes, with a focus on web scraping as an application. The course will combine different domains, i.e., web programming, system programming, and machine learning. In particular, the course focuses on analyzing webpage HTML using Python, extracting available information, and generating dashboards. Co-requisite: COSC 333 (or CCEE 411) and MATH 351 (or BADM 250).		

HCIS 498	Special Topics in Healthcare	1-3 credits
This special topic course will be defined based on current trends and departmental special interests. It may incorporate technical seminars, advanced readings, and special projects as required/approved by the department. **Prerequisite**: Senior standing.		